SaaS Sales, Discount and Profitability

Manju Suthar | Business Analyst portfolio project

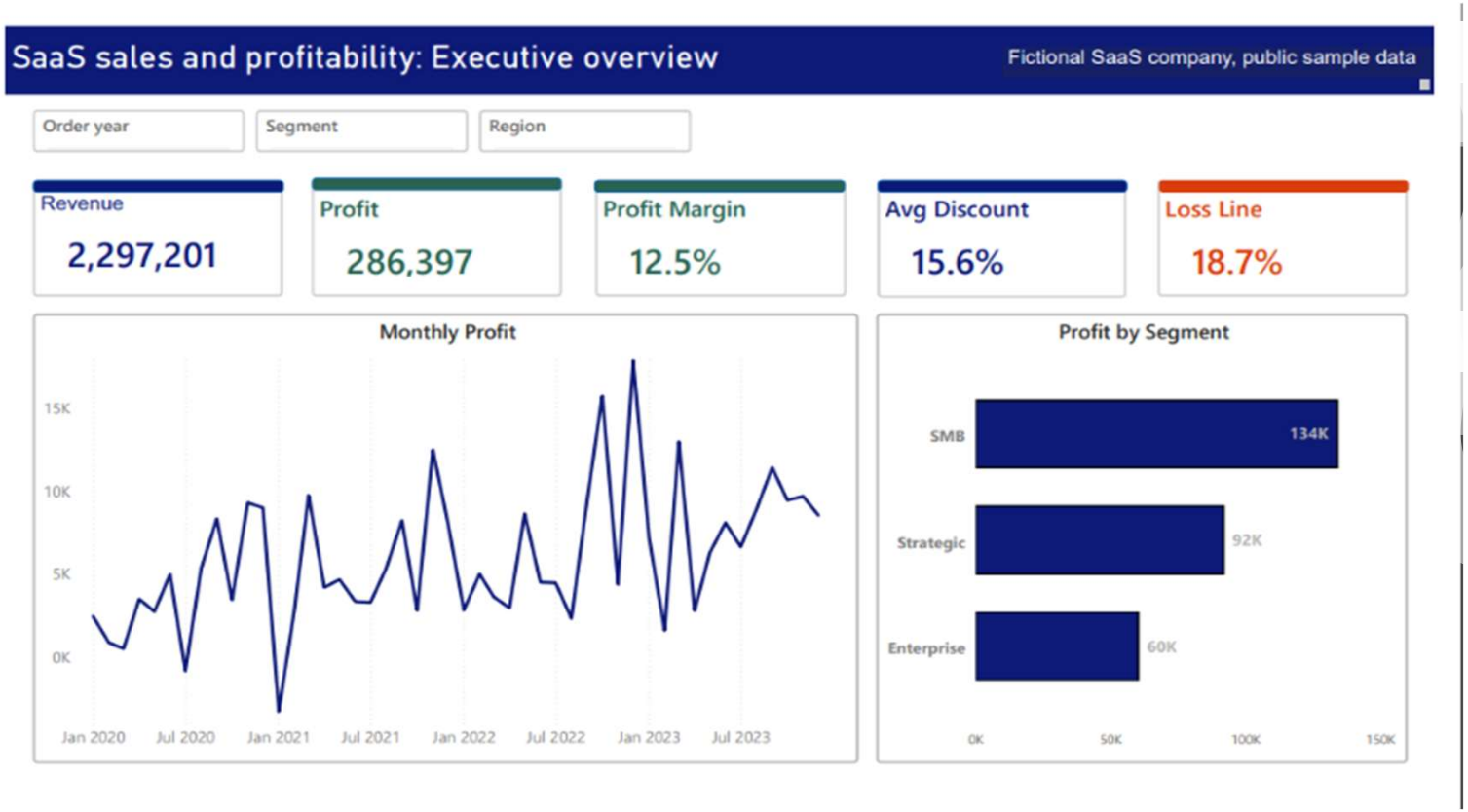
Fictional company, public sample data

Which products and segments make money, and where do discounts destroy profit?

- Data: 9,994 order lines, 2020 to 2023, public sample for a fictional company
- Tools: Excel, SQL, Python, Power BI
- Output: 4-page dashboard, BRD and analysis report

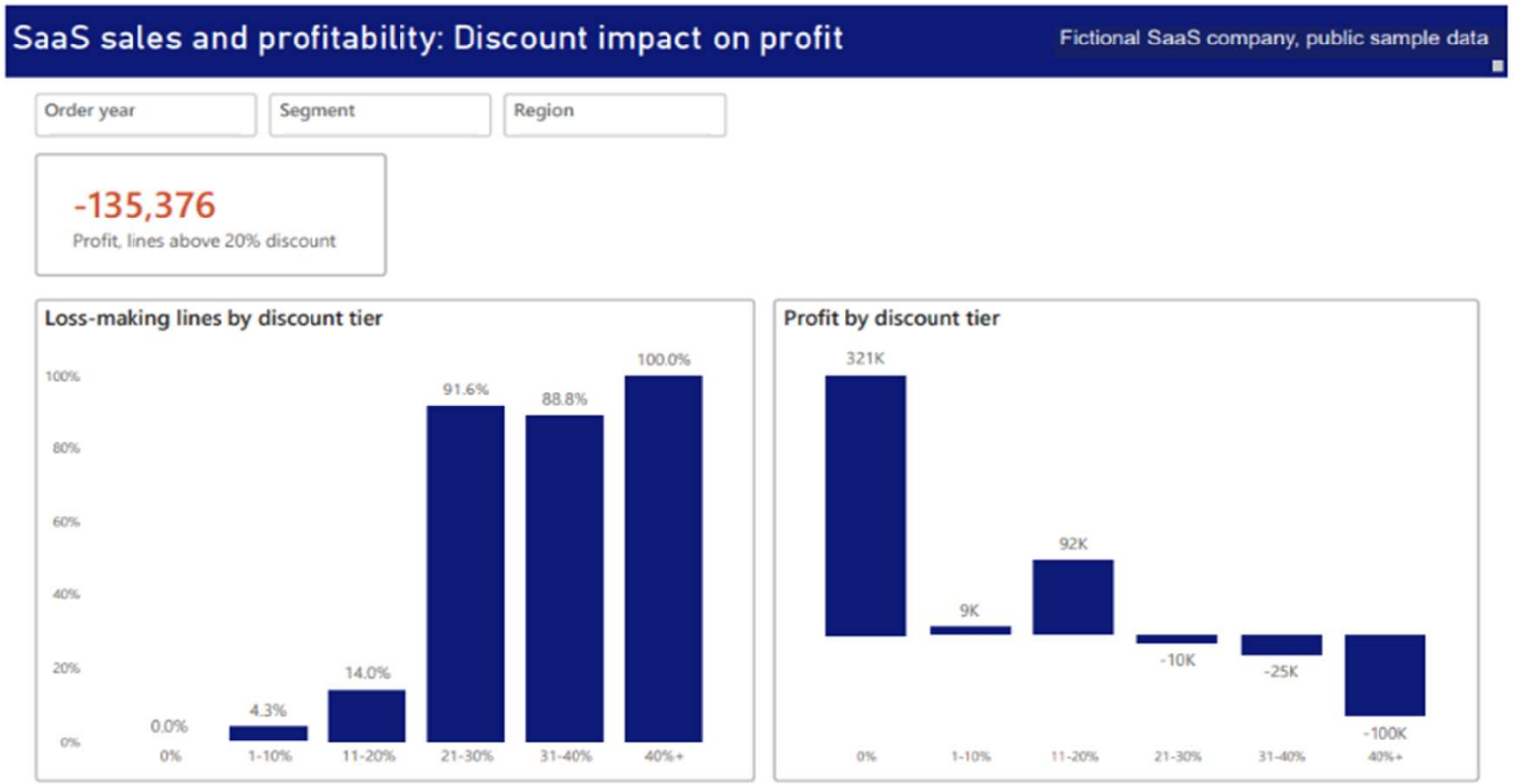
Revenue is 2.3M, but only 12.5% reaches profit

- Revenue 2,297,201 and profit 286,397
- Average discount 15.6%
- 18.7% of order lines lose money



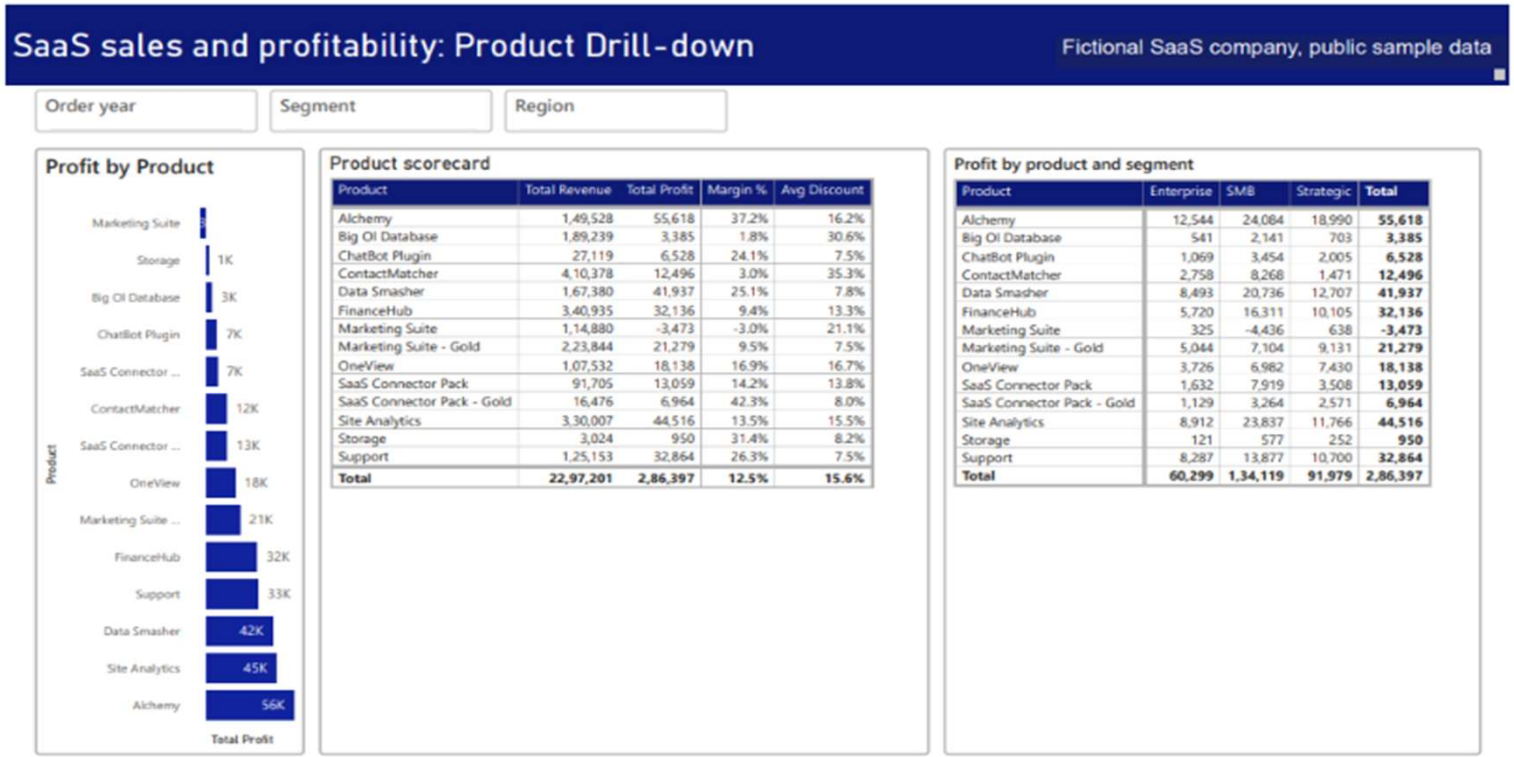
Discounts above 20% lost 135,376

- Loss-making lines: 14.0% at 11-20% discount, 91.6% at 21-30%, 100% at 40%+
- The 0% discount tier earned 321K
- Spearman ρ = -0.54, Kruskal-Wallis p < 0.001



One product loses money; two have thin margins

- Marketing Suite: -3,473 (SMB: -4,436)
- ContactMatcher: 3.0% margin at 35.3% average discount
- Big OI Database: 1.8% margin at 30.6%
- Alchemy earns most: 55,618 (37.2% margin)



Product	Total Revenue	Total Profit	Margin %	Avg Discount
Alchemy	1,49,528	55,618	37.2%	16.2%
Big OI Database	1,89,239	3,385	1.8%	30.6%
ChatBot Plugin	27,119	6,528	24.1%	7.5%
ContactMatcher	4,10,378	12,496	3.0%	35.3%
Data Smasher	1,67,380	41,937	25.1%	7.8%
FinanceHub	3,40,935	32,136	9.4%	13.3%
Marketing Suite	1,14,880	-3,473	-3.0%	21.1%
Marketing Suite - Gold	2,23,844	21,279	9.5%	7.5%
OneView	1,07,532	18,138	16.9%	16.7%
SaaS Connector Pack	91,705	13,059	14.2%	13.8%
SaaS Connector Pack - Gold	16,476	6,964	42.3%	8.0%
Site Analytics	3,30,007	44,516	13.5%	15.5%
Storage	3,024	950	31.4%	8.2%
Support	1,25,153	32,864	26.3%	7.5%
Total	**22,97,201**	**2,86,397**	**12.5%**	**15.6%**

Product	Enterprise	SMB	Strategic	Total
Alchemy	12,544	24,084	18,990	**55,618**
Big OI Database	541	2,141	703	**3,385**
ChatBot Plugin	1,069	3,454	2,005	**6,528**
ContactMatcher	2,758	8,268	1,471	**12,496**
Data Smasher	8,493	20,736	12,707	**41,937**
FinanceHub	5,720	16,311	10,105	**32,136**
Marketing Suite	325	-4,436	638	**-3,473**
Marketing Suite - Gold	5,044	7,104	9,131	**21,279**
OneView	3,726	6,982	7,430	**18,138**
SaaS Connector Pack	1,632	7,919	3,508	**13,059**
SaaS Connector Pack - Gold	1,129	3,264	2,571	**6,964**
Site Analytics	8,912	23,837	11,766	**44,516**
Storage	121	577	252	**950**
Support	8,287	13,877	10,700	**32,864**
Total	**60,299**	**1,34,119**	**91,979**	**2,86,397**

Profit is concentrated, and the bottom 10 customers all lose money

- SMB is 47% of profit
- Top 10 customers earn 77,437 (27%)
- Bottom 10 customers lost 9,381 in total
- 99 customers, 2,893 profit per customer



Recommendations

- Require approval for any discount above 20%
- Scenario: a 20% cap gives profit of about 421,773 to 505,588, against 286,397 today (a scenario, not a forecast)
- Review Marketing Suite pricing and the discounting on ContactMatcher and Big OI Database
- Review terms for the 10 loss-making customers

Method and limits

- Methods: correlation, Kruskal-Wallis test, linear regression ($R^2 = 0.048$), decision tree (splits at 25%)
- Sample data for a fictional company
- Association, not proof of cause
- The scenario assumes customers react as they did in the past

Data: Amazon AWS SaaS Sales Dataset (Kaggle, nnthanh101), fictitious company.